Week 1 Progress

Jack Robbers

Work this week

- Found a major bug with windowing code
- Meant that conclusion about mean being good enough was incorrect
- Made graphs show all the points
- Implemented mean + std metric (and machinery for combining any metrics)
- Got quite close with comparing different window sizes and offsets, will present next week

Bins of size s samples, offset by o samples from the previous bin. o could be less than s.

Profile the data by computing a metric on the data in each bin. Naive approach uses max and min values of each metric across the population.

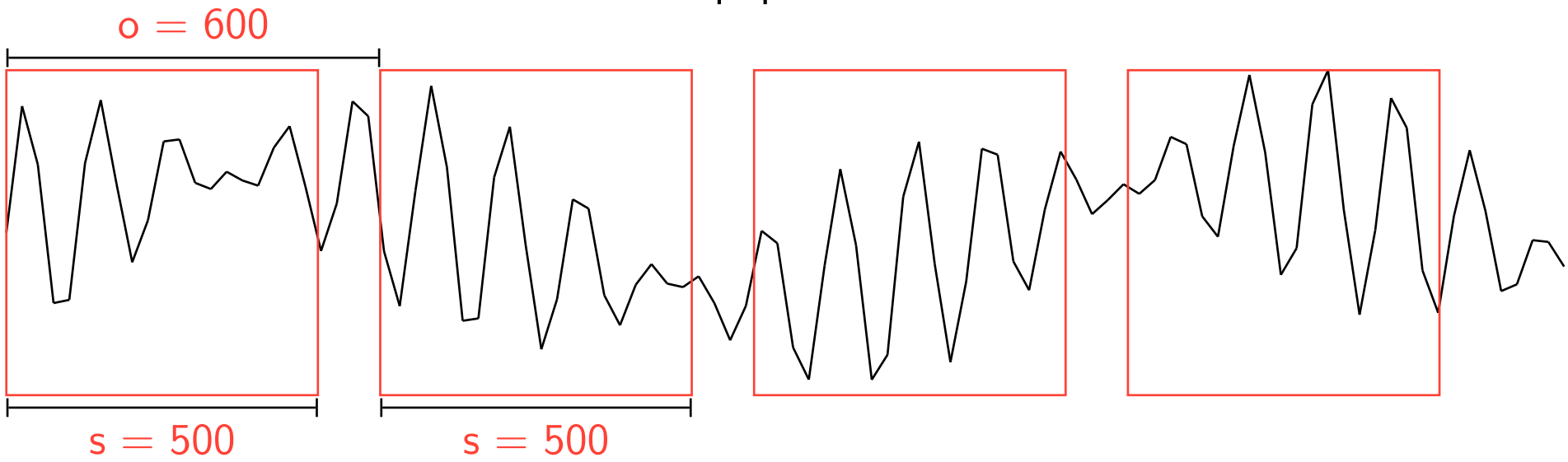
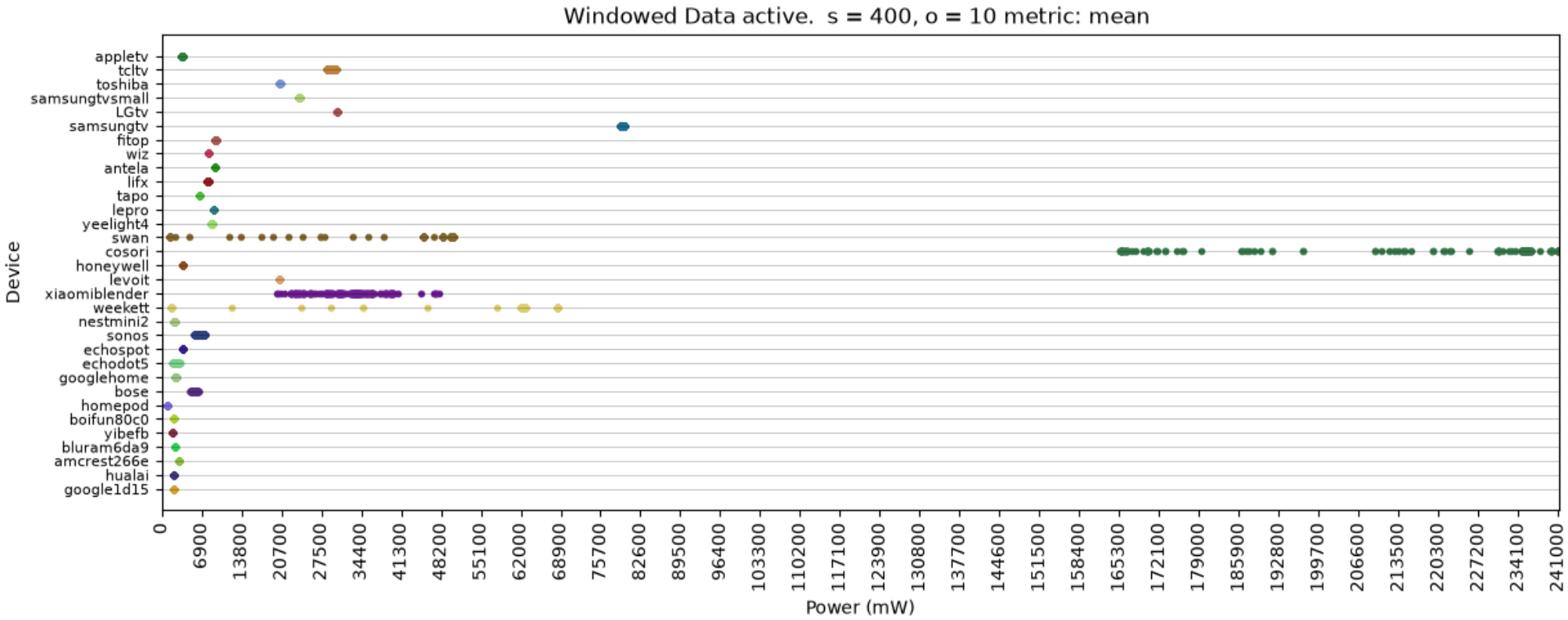
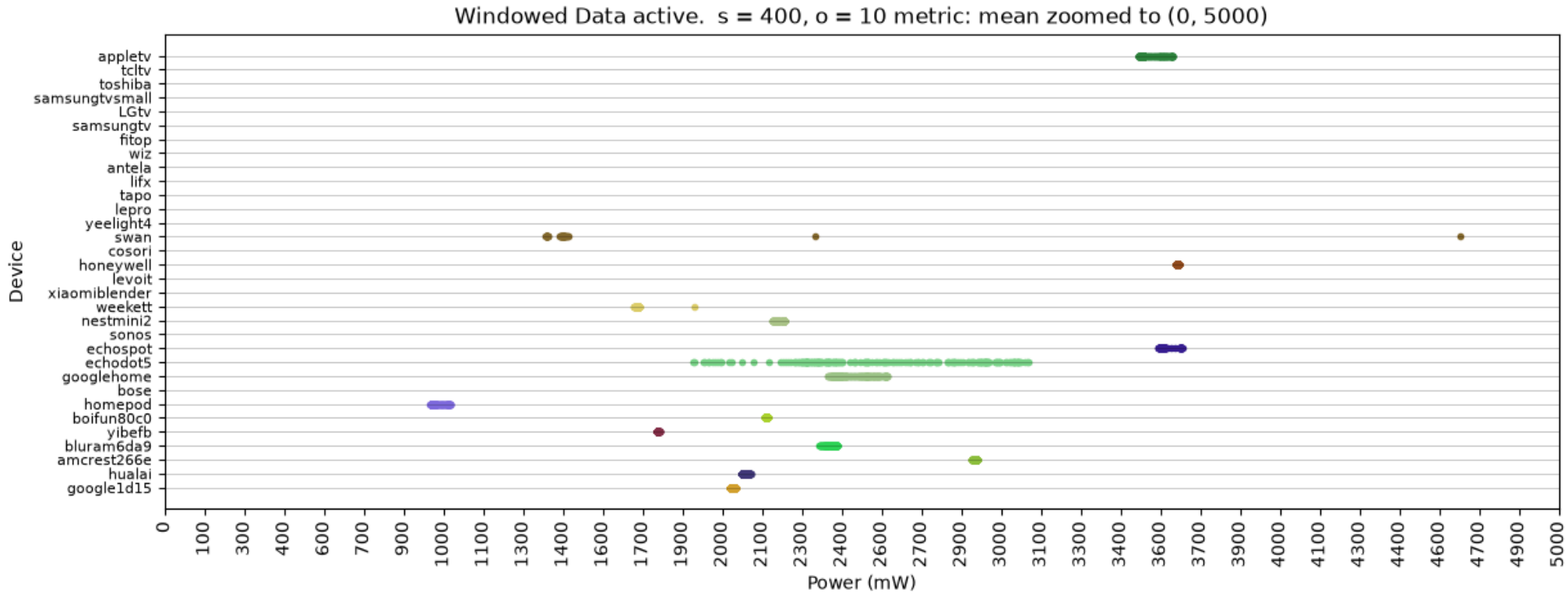


Figure 1: Bins with $o = 600$ and $s = 500$

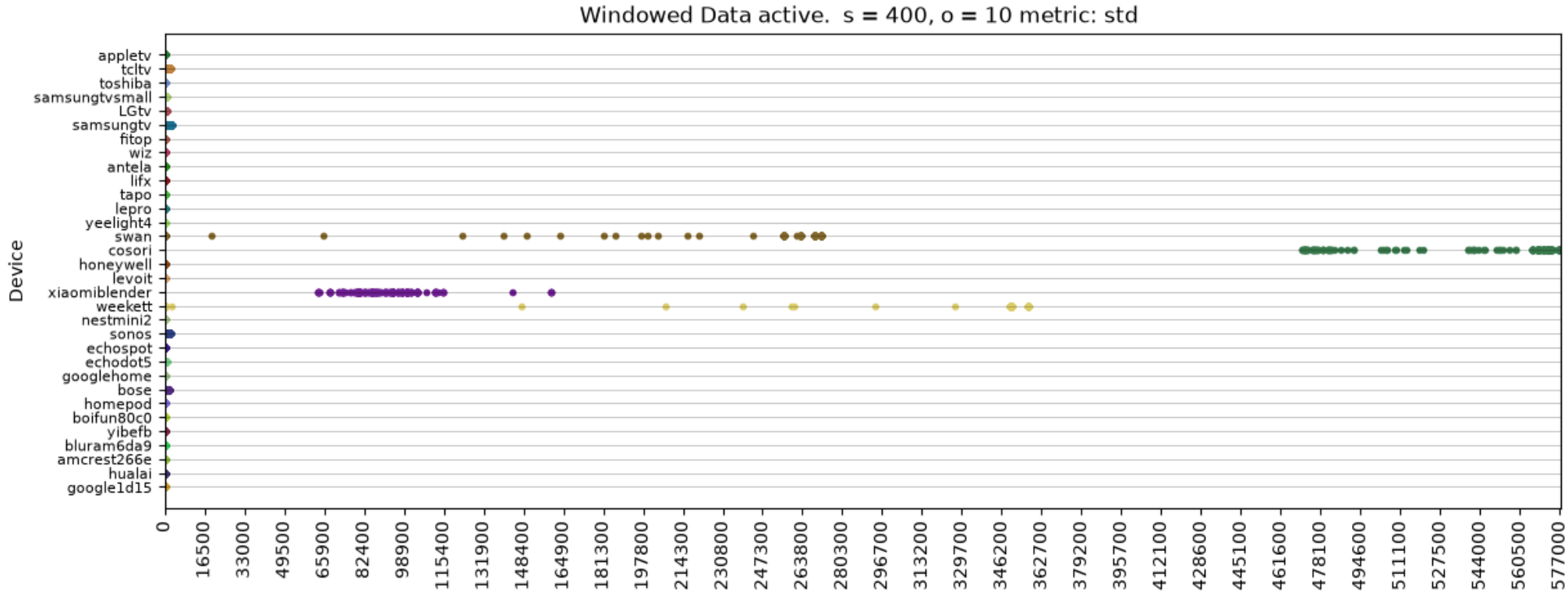
Method 1: by mean



Mean, zoomed in



Method 2: by std



Method 3: Both

TODO: produce 2d graph of this

Evaluation:

Number of fully distinguishable devices:

Method	Idle	Active
Mean	16/32	3/32
Std	0/32	1/32
Both	21/32	3/32

Most devices are low powered, < 50 Watts

Std overlaps more than mean, which makes sense since it doesn't carry information about the actual value, just it's spread.

Test a range of values using colourmesh

Outlier Detection

apply this

Application to unseen data

Blocked by

Confusion Metric

table showing how it should be classified and what actually happened

Conformity Test stats

Highlight some devices that are particularly bad / confusing to each other

Conclusions:

Mean cannot actually fully distinguish this dataset, with the bins set at $s = 400$ and $o = 10$. I had a bug last week

Std cannot, which makes sense since it only measures the spread, no the actual value of result

Next Steps / Limitations

- Implement the absolute overlap metric
- Implement a metric using both
- Different values of o and s. Produce heatmap / colourmesh of this.
- Using only the min and max of a metric means that an outlier can throw off the performance. See if a filter or taking the meidan is better
- Explore fft?